Academic Summary 10 — AT-ADD

AT-ADD: A Benchmark and Challenge for Robust and All-Type Audio Deepfake Detection

Xie et al., ACM Multimedia 2026 Grand Challenge

The paper *AT-ADD: A Benchmark and Challenge for Robust and All-Type Audio Deepfake Detection* by Xie and colleagues (2026) deals with the generalization and robustness limitations of voice forgery detection systems under realistic conditions. The researchers point out that models achieving high performance on controlled evaluation datasets may struggle to detect synthetic speech when they are exposed to new generation systems, to different recording devices, and to signal processing such as compression, noise, and re-recording. In order to examine these challenges, AT-ADD is presented, a Benchmark including two tracks: forgery detection in speech under distribution shifts (Track 1), and forgery detection across a variety of audio types, including speech, environmental sounds, singing, and music (Track 2). The first track is especially relevant to the present study, which focuses on detecting real speech versus synthetic speech.

The Track 1 dataset includes 245,655 recordings, 11,299 identified speakers, 96 base languages, and 47 synthetic-speech generation conditions. The evaluation structure is based on separation between 20 generation systems appearing in the training and development sets and 27 forgery conditions appearing in the evaluation set, including synthetic speech that underwent Replay. In addition, the evaluation data includes diverse recording sources and acoustic environments, alongside changes in signal loudness, speed, and sampling rate. This protocol is intended to examine generalization ability to synthesis systems not seen in training and robustness to conditions that do not fully match the distribution of the development data. However, the combination of several change factors makes it difficult to isolate the contribution of each factor to the decline in performance.

The researchers conducted the challenge within a Closed Setting, in which participants were permitted to use the training and development data provided by the organizers, alongside general pre-trained models and permitted data-augmentation means. The official metric of Track 1 is Macro-F1, computed as the mean of the F1 scores of the Real and Fake classes in order to give equal weight to both. As part of the study, Baseline models based on spectral representations, AASIST, Self-Supervised Learning representations, and Audio LLM models were evaluated. Among the Baselines, the FT-XLSR-AASIST system achieved the highest result on the evaluation set, 76.73% Macro-F1, compared with 60.39% for AASIST. However, even the leading SSL model showed a significant decline relative to its performance on the development set, a fact that illustrates the generalization limitations under distribution shift.

The winning system, WaveShield, achieved 90.71% Macro-F1 by means of a combination of W2V-BERT 2.0 representations, AASIST components and Adapters, staged training strategies, diverse Augmentation, and fusion of the scores of several subsystems. Other leading systems used combinations of XLSR, CQCC, adaptations to channel conditions, audio-segment analysis, and calibration of decision thresholds. These results point to the importance of the overall system — including the audio representation, the training policy, and the inference strategy — and not only the choice of the base architecture. However, in the absence of uniform Ablation experiments, the independent contribution of each component to the performance gaps between the systems cannot be isolated.

An additional central contribution of the paper is error analysis at the level of the generation system and the speech source. The researchers found that BigVGAN and Replay conditions were among the most difficult situations to detect, although most systems achieved high performance on other forgery conditions. At the same time, some of the models wrongly classified a high rate of real recordings from certain sources, in particular 3D-Speaker. These findings emphasize that an overall evaluation score may hide substantial weaknesses, both in detecting fake speech and in preserving the correct identification of real speech. In addition, the analysis of the overlap between the models' errors showed that their failure patterns differ in part, but a simple combination by means of majority voting did not improve the performance of the leading system in Track 1. The Bootstrap check also demonstrated the importance of uncertainty evaluation, as small gaps between systems adjacent in the ranking were not as stable as the winning system's advantage.

The study has several methodological limitations. First, changes in the generation systems, the speech sources, and the recording conditions may occur in parallel, and therefore it is difficult to infer causally what the central factor is for the harm to performance. Second, the researchers report an overlap of 18 speaker identities from Common Voice between the chosen subsets, so that one cannot claim full separation at the speaker level. Third, the analysis of the winning systems is based on binary classification decisions, and not on continuous scores, and therefore does not allow computing EER or ROC analysis from the submissions. Likewise, practical access to the Track 1 dataset has not been verified for the present study, and therefore it should not be assumed that it can be integrated directly into the training or evaluation processes.

In the context of the present project, AT-ADD provides mainly a methodological framework for planning and evaluating voice forgery detectors, and not an active data source for expanding the training. The paper reinforces the need for controlled separation between training, development, and evaluation data, for examining generalization to new synthesis systems, and for analyzing performance by data source and attack type. In addition, it emphasizes the importance of metrics complementary to EER, of uncertainty evaluation, and of controlled experiments for testing the contribution of Augmentation and Fine-Tuning methods.

The central research conclusion is that high performance on development data does not necessarily indicate successful generalization under real-world conditions. A reliable evaluation of voice forgery detectors requires examining distribution shifts, analyzing errors by conditions and sources, and a methodological separation between model selection and its final evaluation.

Reference

Xie, Y., et al. (2026). *AT-ADD: A Benchmark and Challenge for Robust and All-Type Audio Deepfake Detection*. arXiv:2608.23437v2. https://arxiv.org/abs/2608.23437